

TABLE

customer_id	signup_date	last_active_date	subscription_type	monthly_usage_minutes	support_tickets_last_90_days	churned
C001	2023-01-15	2023-07-10	Premium	520.5	2	0
C002	2022-11-03	2023-06-01	Basic	210.0	1	1
C003	2023-03-22	2023-08-15	Free	45.2		1
C004	2021-12-30	2023-07-25	Premium	610.8	3	0
C005	2022-06-18	2023-05-30	Basic	190.4	2	1
C006	2023-02-10	2023-08-20	Free	60.0		0
C007	2022-09-05	2023-07-01	Premium	480.3	1	1
C008	2023-04-01	2023-08-10	Basic	230.7	2	0
C009	2022-10-12	2023-06-15	Free	35.0		1
C010	2023-01-25	2023-08-18	Premium	550.0	4	0

Parsing dates and calculate **days_since_last_active**

All of the dates have been transformed into the American style – first year, then month, then date (2023-11-28) for example (28 of November – year 2023).

1. For C16 cell – I am using this formula: **'=DATEDIF(C16;TODAY();"d")** – the date is 2023-07-10 (last_active_date) – until today (2025-08-23), there are total of **775** days, when you use this formula;
2. For C17 cell – 2023-06-01 (last_active_date) – **'=DATEDIF(C17;TODAY();"d")** – until today (2025-08-23) – total of 814 days;
3. For C18 cell – 2023-08-15 (last_active_date) – **'=DATEDIF(C18;TODAY();"d")** – until today (2025-08-23) – total of 739 days;
4. For C19 cell – 2023-07-25 (last_active_date) – **'=DATEDIF(C19;TODAY();"d")** – until today (2025-08-23) – total of 760 days;
5. For C20 cell – 2023-05-30 (last_active_date) – **'=DATEDIF(C20;TODAY();"d")** – until today (2025-08-23) – total of 816 days;
6. For C21 cell – 2023-08-20 (last_active_date) – **'=DATEDIF(C21;TODAY();"d")** – until today (2025-08-23) – total of 734 days;
7. For C22 cell – 2023-07-01 (last_active_date) – **'=DATEDIF(C22;TODAY();"d")** – until today (2025-08-23) – total of 784 days;
8. For C23 cell – 2023-08-10 (last_active_date) – **'=DATEDIF(C23;TODAY();"d")** – until today (2025-08-23) – total of 744 days;
9. For C24 cell – 2023-06-15 (last_active_date) – **'=DATEDIF(C24;TODAY();"d")** – until today (2025-08-23) – total of 800 days;
10. For C25 cell – 2023-08-19 (last_active_date) – **'=DATEDIF(C25;TODAY();"d")** – until today (2025-08-23) – total of 736 days.

That was for each of the customer's last active date – how many days (in number) have passed until today.

Normalizing **monthly_usage_minutes** using min-max scaling.

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So, we are going back to the table again. I am going to calculate it as 30 days of the month – 1 day = 1440 minutes, 30 days * 1440 = 43200 minutes.

- 520.5 minutes / 43200 minutes (30 days total) = 0,0120486111111111 * 100 = 1,204861111111111 or ~ **1,20% of the total month; (E16 cell)**
- 210.0 minutes / 43200 minutes (30 days total) = 0,0048611111111111 * 100 = 0,486111111111111 or ~ **0,49 % of the total month; (E17 cell)**
- 45.2 minutes / 43200 minutes (30 days total) = 0,0010462962962963 * 100 = 0,1046296296296296 or ~ **0.10 % of the total month; (E18 cell)**
- 610.8 minutes / 43200 minutes (30 days total) = 0,0141388888888889 * 100 = 1,413888888888889 or ~ **1.41% of the total month; (E19 cell)**
- 190.4 minutes / 43200 minutes (30 days total) = 0,0044074074074074 * 100 = 0,4407407407407407 or ~ **0.44% of the total month; (E20 cell)**
- 60.0 minutes / 43200 minutes (30 days total) = 0,0013888888888889 * 100 = 0,138888888888889 or ~ **0.14 % of the total month; (E21 cell)**
- 480.3 minutes / 43200 minutes (30 days total) = 0,0111805555555556 * 100 = 1,118055555555556 or ~ **1.11 % of the total month; (E22 cell)**
- 230.7 minutes / 43200 minutes (30 days total) = 0,0053402777777778 * 100 = 0,534027777777778 or ~ **0.53 % of the total month; (E23 cell)**
- 35.0 minutes / 43200 minutes (30 days total) = 0,000810185 * 100 = 0,0810185 or ~ **0,08% of the total month; (E24 cell)**
- 550.0 minutes / 43200 minutes (30 days total) = 0,0127314814814815 * 100 = 1,273148148148148 or ~ **1.27% of the total month (E25 cell)**

Conclusion – for example 520.5 minutes (520m and 30s) is ~ 1.20% of 43200 minutes (1 month of 30 days).

Filling missing values in support_tickets_last_90_days with median grouped by subscription_type

So, I am sharing the table again, for the third time:

customer_id	signup_date	last_active_date	subscription_type	monthly_usage_minutes	support_tickets_last_90_days	churned
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We have these numbers – 2 (F16 cell), 1 (F17 cell), Empty (F18 cell), 3 (F19 cell), 2 (F20 cell), Empty (F21 cell), 1 (F22 cell), 2 (F23 cell), Empty (F24 cell), 4 (F25 cell).

So, how to calculate using the median way – to get the write answer to fill the empty cells – the answer is 2 and I will explain now how is 2:

We have numbers – 2, 1, 3, 2, 1, 2 and 4. (total of 7 numbers) We change their positions to ascending way, which means – 1, 1, 2, 2, 2, 3, 4. Using the median technique, the number in the center is the answer. We have 7 numbers and in the center is the 4th number – 2. The correct answer is 2. So, I am filling the empty cells now:

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And that's it. Going to the next task:

Creating a pivot table showing average usage and ticket count per subscription_type

Row Labels	Average of monthly_usage_minutes	Sum of support_tickets_last_90_days	Count of support_tickets_last_90_days
Basic	#DIV/0!	5	3
Free	#DIV/0!	6	3
Premium	#DIV/0!	10	4
Grand Total	#DIV/0!	21	10

Unfortunately, there is a problem with the pivot table, so I am going to calculate it manually:

For Basic Subscription Type: $210.0 + 190.4 + 230.7 = 631.1 / 3 = 210.3666666666667$ or ~ **210.37** of monthly usage minutes;

For Free Subscription Type: $45.2 + 60.0 + 35.0 = 140.2 / 3 = 46.73333333333333$ or ~ **46,73** of monthly usage minutes;

For Premium Subscription Type: $520.5 + 610.8 + 480.3 + 550.0 = 2161.6 / 4 = 540.4$ of monthly usage minutes.

Count of tickets per subscription type: Basic – 3, Free – 3, Premium – 4 = **10** (total)

Identify top 3 features that correlate most with churned

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Churned <> Subscription Type

So, about the churned customers there are total of 5:

- Basic Subscription Type – 2;
- Free Subscription Type – 2;
- Premium Subscription Type – 1;

That means that it is more likely for the Basic and Free Subscription Type users to churn.

Churned <> Support Tickets Last 90 Days:

The churned users support 1 or 2 tickets and not more than that. The users that support more than 2 do not churn, which means that the more support the less likely is to drop out.

Churned <> Monthly Usage Minutes

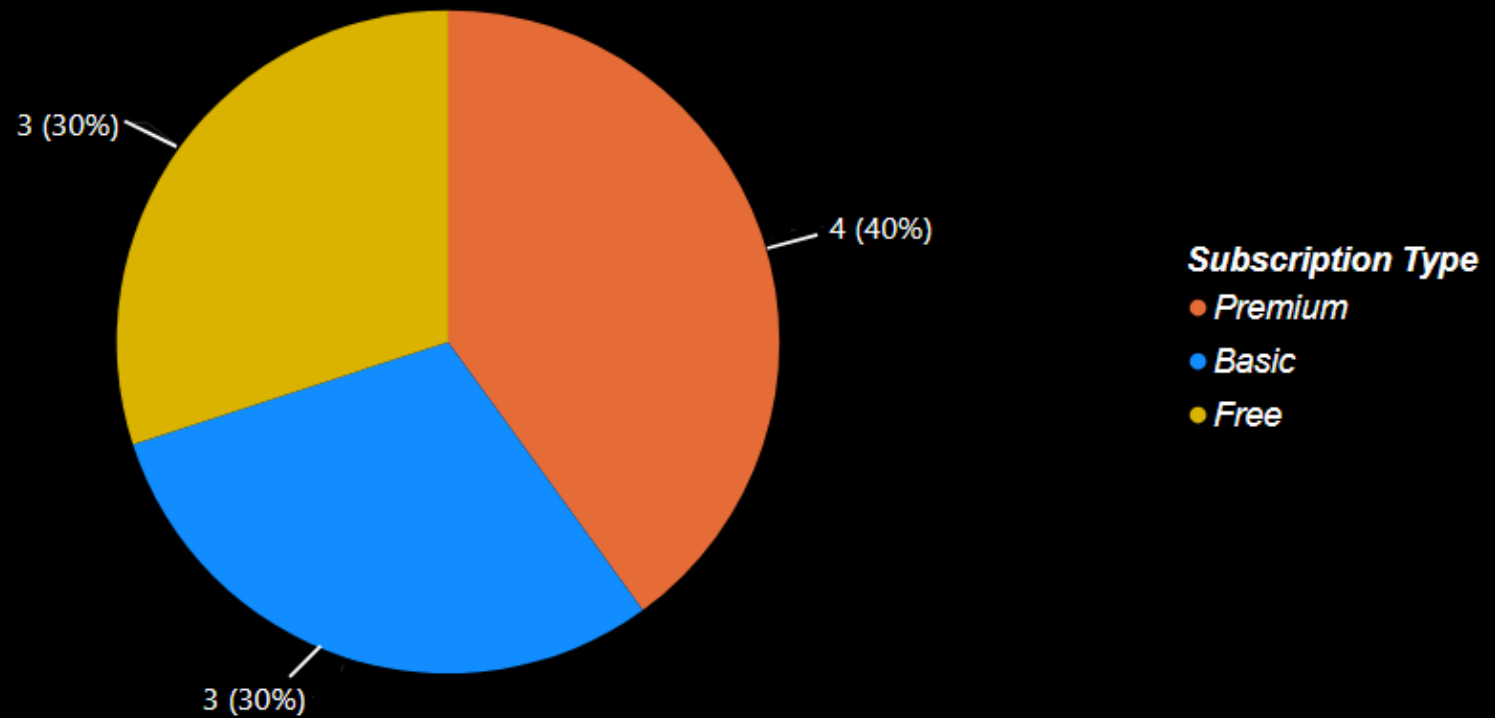
Something interesting here is that the customers with the top minutes (more than 500) of monthly usage do not churn (520.5, 550.0 and 610.80). Also those users with the lowest time usage monthly are churned (35.0 and 45.2). That proves that it is very likely the users with very low minutes of usage to go away and the higher you have the less likely is to drop out.

Well, there are some exceptions, of course. There is one person (C007) with 480.3 minutes of usage, but there is only 1 ticket support. Same goes for user C002 with 210.0. He churns with just 1 ticket support. Another interesting moment is that those with 1 support with average level of monthly usage churn out.

So even if you have good level of usage if you do not support much that means that you obviously have problems and complains about the service.

Customers based on Subscription

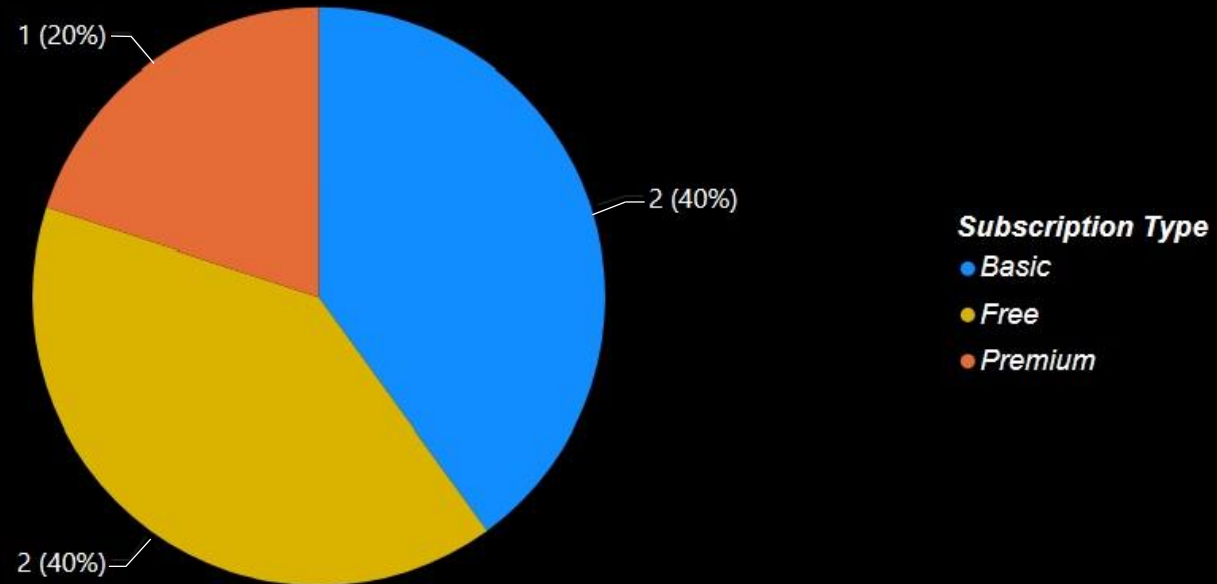
10 people participating in the analysis. How many of them are pulling off?



So, we have total of 10 people. Each of them is using a type of subscription. And how many of them are complaining and churn out?

Churned customers based on Subscription

How many of the users have complains? And what type of subscription they are using?



To be a premium user is to be a pleased customer. Well, not all of them are pleased enough. Still, the important is for the business to have the ability and solve problems when there are. Because, the best service is the service with a true strategy. Well, it looks like that according to our analysis there are some people from each of the subscription types that is not happy enough. We have 2 people of 3 using basic type to go away. With free type there are 2 people as well of 3. And 1 premium user of 4 has dropped out.

It is the most important thing to periodically check from problems, read customer complains if there are, take care of the material resources to be with high quality and of course last but not least the workers. If you have a great care about all that, then you can beat all the competition and have the happiest clients ever.

This report was created by:



NIKOLAY VETSOV

DATA ANALYSIS INTERN AND AI ENTHUSIAST

Tools used for this project:

Microsoft Word, Microsoft Excel, Microsoft Publisher, Microsoft Power BI, Paint 3D, AI (for explaining in complicated moments)

That was combined project – related to Data Analysis and Data Engineering

Thank you for your patience!